power — probe header and test points

One probe position for every board: the 1×8 header J_PROBE at the top left. Pin 1 is GND.

J_PROBE (1×8, row 1, columns 6–13)

Pin	Label	Net	Install
1	GND	GND	CP09
2	5V	5V_SYS	CP09
3	3V3	3V3_SYS	CP09
4	12V	12V_SYS	CP09
5	SIG1	BUCK_5V	CP09
6	SIG2	BUCK_3V3	CP09
7	TRIG	EDU_PWR_EN	CP09
8	FAULT	PRESET_N	CP09

Scope ground clip on pin 1 (GND), probe tip on the pin you need. TRIG is a convenient scope trigger; FAULT shows the latched fault or its reset line.

Designated test pads

Net	Pad	Column, row	Install
GND	J_PROBE.1	6, 1	CP09
GND	U_POWER_INTERLOCK_ALLOW.13	73, 6	CP09
12V_FUSED	J_POWER_INPUT_SW_MAIN.1	30, 35	CP09
BUCK_5V	J_PROBE.5	10, 1	CP09
BUCK_3V3	J_PROBE.6	11, 1	CP09
12V_SYS	J_PROBE.4	9, 1	CP09
5V_SYS	J_PROBE.2	7, 1	CP09
3V3_SYS	J_PROBE.3	8, 1	CP09
5V_EDU	J_POWER_INTERLOCK_MATRIX.1	45, 5	CP09
EDU_PWR_EN	J_PROBE.7	12, 1	CP09
ALLOW_RAIL	JP_R7_PASS.1	68, 34	CP11
NOT_FAULT	U_POWER_INTERLOCK_ALLOW.2	70, 6	CP11
PRESET_N	J_PROBE.8	13, 1	CP09
ROUTER_ALIVE	J_POWER_INTERLOCK_MATRIX.9	49, 5	CP09

Analogue nodes are not brought out. Reference, current-sense and gate-drive nets stay short; probe them at the part pad, or use the existing J_POWER_INTERLOCK_TEST header on the power board.

router-matrix — probe header and test points

One probe position for every board: the 1×8 header J_PROBE at the top left. Pin 1 is GND.

J_PROBE (1×8, row 1, columns 6–13)

Pin	Label	Net	Install
1	GND	GND	CP00
2	5V	5V_SYS	CP00
3	3V3	USB_ROUTER_3V3	CP00
4	12V	12V_SYS	CP00
5	SIG1	ROUTER_ALIVE	CP00
6	SIG2	HEARTBEAT	CP00
7	TRIG	XBAR_LATCH	CP00
8	FAULT	EDU_FAULT	CP00

Scope ground clip on pin 1 (GND), probe tip on the pin you need. TRIG is a convenient scope trigger; FAULT shows the latched fault or its reset line.

Designated test pads

Net	Pad	Column, row	Install
GND	J_PROBE.1	6, 1	CP00
GND	U_MATRIX_CONTROL_DIAG.5	75, 7	CP00
USB_ROUTER_3V3	J_PROBE.3	8, 1	CP00
5V_SYS	J_PROBE.2	7, 1	CP00
12V_SYS	J_PROBE.4	9, 1	CP00
EDU_PWR_EN	J_ROUTER_CORE_PROTOTYPE.4	20, 20	CP00
HEARTBEAT	J_PROBE.6	11, 1	CP00
ROUTER_ALIVE	J_PROBE.5	10, 1	CP00
EDU_FAULT	J_PROBE.8	13, 1	CP00
XBAR_DAT	U_ROUTER_CORE_ESP1.21	14, 6	CP00
XBAR_CLK	U_ROUTER_CORE_ESP1.28	14, 13	CP00
XBAR_LATCH	J_PROBE.7	12, 1	CP00
STB_A_IN	U_ROUTER_CORE_ESP1.27	14, 12	CP00
STB_B_IN	U_ROUTER_CORE_ESP1.25	14, 10	CP00
STROBE_A	JP_R7_STBA.1	46, 33	CP24
STROBE_B	JP_R7_STBB.1	50, 33	CP24
XBAR_RESET	U_MATRIX_SWITCHES_MT_A.3	38, 6	CP24
RST_RELEASE	JP_R7_RST.1	8, 35	CP00
FIX_SDA	J_ROUTER_CORE_FIXTURE.3	6, 27	CP00
FIX_SCL	J_ROUTER_CORE_FIXTURE.4	7, 27	CP00

Analogue nodes are not brought out. Reference, current-sense and gate-drive nets stay short; probe them at the part pad, or use the existing J_POWER_INTERLOCK_TEST header on the power board.